

Amirkabir University of Technology
Tehran Polytechnic
Department of Mathematics and Computer Science

Computer Simulation Course - Project 2

Rare Event Estimation in Product Quality Control

Using Real Data

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Introduction and Project Description

In the real world of engineering and quality management, the occurrence of a catastrophic error or a severe drop in the quality of a product is considered a "rare event". Although the probability of these events is very low, in the event of occurrence, they carry heavy financial and reputational costs. The goal of this project is to use computer simulation methods to model a real process, generate data corresponding to reality, and finally accurately estimate the probability of these rare and critical events.

General Problem Statement

You must choose a real dataset that contains a continuous variable (such as quality, strength, purity percentage, etc.). Then, investigate what value of this variable is identified as a "critical failure" (e.g., falling in the 1% extreme tail of the distribution). To estimate the probability of this rare event, you cannot solely rely on built-in functions; instead, you must model the behavior of the data using the Accept-Reject algorithm and then, using Standard Monte Carlo and Importance Sampling techniques, estimate the probability of this event with the lowest possible variance.

Suggested Datasets

Students are required to choose one of the following three datasets (or a similar one approved by the instructor) for their project:

- **Wine Quality Dataset:** Available on Kaggle or UCI. The objective is to investigate the probability of producing a wine with extremely high volatile acidity that leads to the return of the entire shipment.
- **Water Quality Dataset:** Available on Kaggle. The objective is to model the distribution of pH or Chloramines in water and estimate the probability of these values exiting the safe and standard range, which is a rare but dangerous event.
- **Concrete Compressive Strength:** Available on UCI. The objective is to simulate the compressive strength distribution of produced concretes and calculate the probability that a concrete sample has a strength lower than the critical threshold (building collapse).

Project Requirements

Your project must include the following stages:

(1) Data Generation with Accept-Reject Algorithm

The true distribution of the target variable in your data is definitely not normal and standard.

- Plot the histogram of the real data.
- Fit an analytical distribution (such as Gamma, Weibull, or a mixture) onto it.
- Using the Accept-Reject algorithm and an envelope distribution (like Normal with high variance or Laplace), generate new random data that follow this distribution.
- Plot the graph of the generated data over the original data histogram and analyze the efficiency of the algorithm.

(2) Standard Monte Carlo Simulation

Using the samples generated in the previous stage:

- Estimate the probability of the rare event occurrence (e.g., quality drop below the allowable limit).
- Perform this task for four distinct sample sizes (e.g., $N = 10^3, 10^4, 10^5, 10^6$).
- Calculate the absolute error and simulation variance, and demonstrate why the Standard Monte Carlo method is inefficient for rare events.

(3) Importance Sampling

To solve the high variance problem in rare event estimation:

- Select an appropriate proposal distribution and shift it towards the critical region (rare event), and justify your choice using mathematical relations.
- Perform the simulation from this new distribution and calculate the exact probability by applying the importance weights.
- Analyze the error reduction behavior and compare the results with Standard Monte Carlo (previous section).

Analytical Questions

1. How does changing the envelope distribution in the Accept-Reject algorithm affect the acceptance rate? Analyze the relationship of this rate with computational time.

2. In Importance Sampling, why does choosing a proposal distribution that has "heavier tails" than the target distribution reduce the risk of variance instability?
3. By plotting the Log-Log graph for the absolute error versus N , compare the slope of the graph in the two methods of Standard Monte Carlo and Importance Sampling. What does this slope indicate about the convergence rate?
4. Under what conditions (in terms of distribution shape or threshold value) might the Importance Sampling method have a weaker performance compared to Standard Monte Carlo?

Notes and Final Report

Constraints and Notes

- The use of Python for all stages including data processing, sample generation, simulation, and plotting is mandatory.
- In the data generation section, using built-in Python functions to directly extract samples from the target distribution is not allowed; the Accept-Reject algorithm must be coded from scratch (base level).
- The choice of the proposal distribution in the Importance Sampling method should not be random; rather, it must be justified with mathematical reasoning and representation in the report.
- The comparison of variance and absolute error between the two methods (Monte Carlo and Importance Sampling) must be clearly presented through tables and error convergence graphs (relative to the increase in N).

Suggested Report Format

Your final report is better to include the following sections:

1. **Introduction and problem statement:** Introducing the dataset, selected feature, and exact determination of the rare event threshold.
2. **Statistical analysis and distribution fitting:** Plotting the real data histogram and selecting the best fitted distribution.
3. **Accept-Reject algorithm implementation:** Introducing the envelope distribution, calculating the scaling factor, and analyzing the acceptance rate.
4. **Standard Monte Carlo simulation:** Executing the simulation, calculating the probability, and analyzing the high variance problem.
5. **Importance Sampling implementation:** Introducing the proposal distribution, calculating the importance weights, and re-estimating the probability.
6. **Evaluation and results comparison:** Analyzing error convergence behavior, comparing the efficiency of the two methods, and interpreting the graphs.
7. **Final conclusion.**